Task 7: Course Registration System

Use Cases, Descriptions, Test Cases, and Diagram

This work is based on the UniPlan SRS. It lists the system actors and use cases, describes each use case, and includes test cases based on their flows.

1. Scope and actors

The MVP is a standalone web application with two user roles: Student and Admin. It uses prepared sample academic data and does not integrate with a live university system. The system boundary therefore has no external university actor.

Actor	What the actor does
Student	Creates an account, views personal academic data and study plan, searches courses, registers or drops courses, and views schedule, progress, and registration history.
Admin	Manages courses, the study plan, sections, semester dates, registration and drop periods, GPA credit rules, and registration records.

Admin account creation is required to be secure and not public, but the SRS leaves the exact provisioning method open. This analysis does not invent an extra user role for that process.

2. Use case list

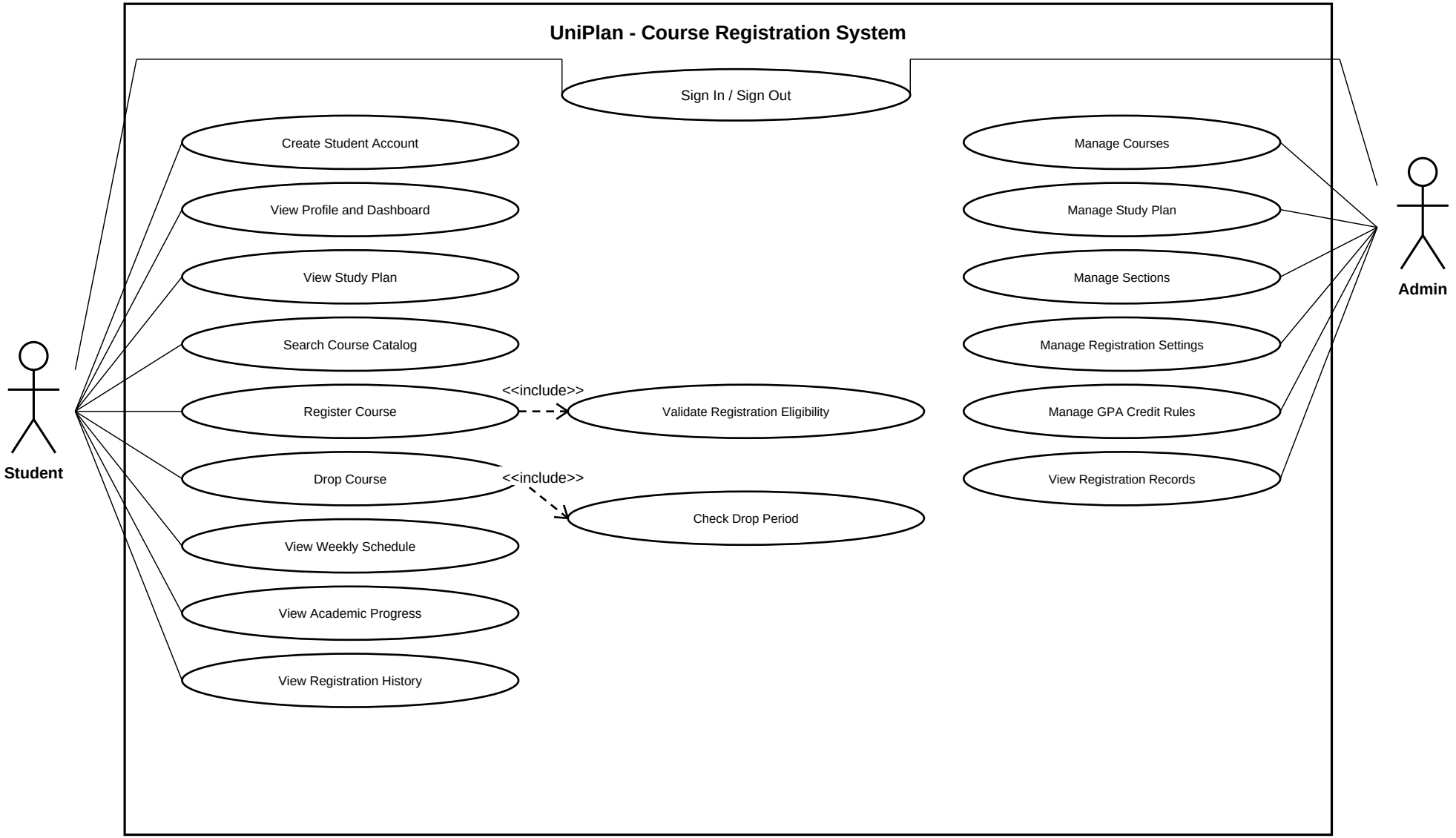
Student: UC-01 Create Student Account; UC-02 Sign In and Sign Out; UC-03 View Profile and Dashboard; UC-04 View Study Plan; UC-05 Search Course Catalog; UC-06 Register Course; UC-07 Drop Course; UC-08 View Weekly Schedule; UC-09 View Academic Progress; UC-10 View Registration History.

Admin: UC-11 Manage Courses; UC-12 Manage Study Plan; UC-13 Manage Sections; UC-14 Manage Semester and Registration Settings; UC-15 Manage GPA Credit Rules; UC-16 View Registration Records.

Included system use cases: UC-17 Validate Registration Eligibility; UC-18 Check Drop Period.

3. Use Case Diagram

Register Course includes server-side eligibility validation. Drop Course includes checking the permitted drop period.



4. Use Case Descriptions

The flow steps below describe the normal path. Alternative flows come from rejection and exception rules stated in the SRS.

UC-01 - Create Student Account

Primary actor: Student	**Requirements:** FR-AUTH-01

Precondition: The student opens the public student sign-up page.

Main flow:
1. The student enters the requested account information.
2. The system validates the submitted information and creates the student account.
3. The system confirms that the account was created.

Alternative flows:
- If required information is missing or invalid, the system shows a validation message and does not create the account.

Postcondition: A student account exists.

UC-02 - Sign In and Sign Out

Primary actor: Student, Admin	**Requirements:** FR-AUTH-02 to 06

Precondition: The user has an account.

Main flow:
1. The user enters their credentials and submits them.
2. The system authenticates the user and applies the user's role permissions.
3. The user can sign out; the system ends the session.

Alternative flows:
- If authentication fails, access is not granted.
- A student cannot open admin operations or another student's data.

Postcondition: The user is signed in with access limited to their role and ownership.

UC-03 - View Profile and Dashboard

Primary actor: Student	**Requirements:** FR-PRO-01 to 07, FR-AUTH-06

Precondition: The student is signed in.

Main flow:
1. The student opens their dashboard.
2. The system shows the student's basic data, program, GPA, passed and registered courses, passed hours, plan hours, remaining hours, registered course count, available registration hours, and plan progress.
3. The student uses shortcuts to open the plan, course list, or schedule.

Alternative flows:
- A request for another student's data is denied.

Postcondition: The student sees their own current academic summary.

UC-04 - View Study Plan

Primary actor: Student	**Requirements:** FR-PLAN-01 to 04

Precondition: The student is signed in and has a study plan.

Main flow:
1. The student opens the study plan.
2. The system shows plan courses arranged by semester and labels each course with its status.
3. The student selects a locked course and the system shows the reason it is locked.

Alternative flows:
- A course not offered in the current semester is marked Not Offered.

Postcondition: The student can identify completed, registered, available, locked, and not-offered courses.

UC-05 - Search Course Catalog

Primary actor: Student	**Requirements:** FR-CAT-01 to 06

Precondition: The student is signed in and current-semester course data is available.

Main flow:
1. The student searches by course name or code.
2. The system lists matching courses offered in the current semester.
3. The system shows each course's name, code, credit hours, prerequisites, sections, lecture times, and remaining seats.

Alternative flows:
- If there are no matches, the system shows an empty result message.

Postcondition: The student can review course and section choices.

UC-06 - Register Course

Primary actor: Student	**Requirements:** FR-REG-01 to 04, 06, 08; FR-VAL-01 to 09; FR-CRD-01 to 05; FR-SCH-04; BR-01 to 06, 10, 11

Precondition: The student is signed in, registration is open, and the selected course and section are offered.

Main flow:
1. The student selects one available course and section.
2. The system shows course details, credit hours, and lecture times.
3. The student confirms the request.
4. The backend validates offering, registration dates, prerequisites, duplicate registration, section seats, GPA-based credit limit, and schedule conflicts.
5. If every check passes, the system records the registration, updates the student's course list, credit-hour total, and schedule, and confirms the result.

Alternative flows:
- If any validation fails, the backend blocks the request and explains the reason.
- Only one course is registered per operation.
- If concurrent requests compete for the last seat, section capacity is not exceeded.

Postcondition: The registration is recorded only after all backend checks pass.

UC-07 - Drop Course

Primary actor: Student	**Requirements:** FR-REG-05 to 07; BR-07

Precondition: The student is signed in and has an active course registration.

Main flow:
1. The student selects a registered course and requests to drop it.
2. The system checks that the permitted drop period is open.
3. The system changes the registration status, records the drop date, and updates the student's course list, credit-hour count, and schedule.

Alternative flows:
- If the drop period is closed, the system rejects the request and explains why.

Postcondition: The registration history is retained; the course is not permanently erased.

UC-08 - View Weekly Schedule

Primary actor: Student	**Requirements:** FR-SCH-01 to 04

Precondition: The student is signed in.

Main flow:
1. The student opens the weekly schedule.
2. The system displays registered course names, lecture days, start and end times, and location when available.
3. After a successful registration or drop, the schedule reflects the updated courses.

Alternative flows:
- If no courses are registered, the schedule shows no scheduled classes.

Postcondition: The student sees the latest weekly schedule.

UC-09 - View Academic Progress

Primary actor: Student	**Requirements:** FR-PRG-01 to 06, FR-PRO-07

Precondition: The student is signed in and study-plan data is available.

Main flow:
1. The student opens academic progress.
2. The system shows required plan hours, earned hours, remaining hours, completed and not-yet-passed courses, and progress percentage.
3. The system counts passed courses only as earned hours.

Alternative flows:
- Currently registered courses are not counted as earned hours.

Postcondition: Progress values reflect passed courses and plan requirements.

UC-10 - View Registration History

Primary actor: Student	**Requirements:** FR-REG-09

Precondition: The student is signed in.

Main flow:
1. The student opens registration history.
2. The system lists course, section, semester, registration date, current status, and drop date when applicable.

Alternative flows:
- If no history exists, the system shows an empty history.

Postcondition: The student can review current and past registration records.

UC-11 - Manage Courses

Primary actor: Admin	**Requirements:** FR-ADM-01 to 03, 14, 15

Precondition: The admin is signed in and authorized.

Main flow:
1. The admin adds or edits course data, including code, credit hours, and prerequisites.
2. The system validates the course data.
3. The admin can disable a course that is no longer offered.
4. The system keeps a course linked to historical registrations.

Alternative flows:
- Duplicate course codes are rejected.
- A course cannot be its own prerequisite.
- Existing registration records are not corrupted by course changes.

Postcondition: Course catalog data is updated consistently.

UC-12 - Manage Study Plan

Primary actor: Admin	**Requirements:** FR-ADM-04 to 06

Precondition: The admin is signed in and authorized.

Main flow:
1. The admin chooses a study plan to edit.
2. The admin assigns courses to the plan and distributes them across semesters.
3. The admin updates the total credit hours required and saves.

Alternative flows:
- The system keeps the plan data organized and reflects saved changes.

Postcondition: The study plan contains the updated course and hour information.

UC-13 - Manage Sections

Primary actor: Admin	**Requirements:** FR-ADM-07 to 10, 16

Precondition: The admin is signed in and a course and semester exist.

Main flow:
1. The admin creates or edits a section, including capacity and lecture times.
2. The system updates registered-student and remaining-seat information.
3. The admin can close a section to new registrations.

Alternative flows:
- The system rejects a capacity reduction below the number of already registered students.

Postcondition: Section details and availability are updated without exceeding capacity.

UC-14 - Manage Semester and Registration Settings

Primary actor: Admin	**Requirements:** FR-ADM-11 to 13

Precondition: The admin is signed in and authorized.

Main flow:
1. The admin sets the current semester.
2. The admin sets registration start and end dates.
3. The admin sets the permitted course-drop period and saves.

Alternative flows:
- Student registration and drop requests are accepted or rejected according to the saved dates.

Postcondition: The current semester and registration windows are configured.

UC-15 - Manage GPA Credit Rules

Primary actor: Admin	**Requirements:** FR-CRD-01 to 05; FR-ADM-17

Precondition: The admin is signed in and authorized.

Main flow:
1. The admin opens credit-hour rules.
2. The admin edits GPA ranges and their maximum credit hours.
3. The system validates that ranges do not overlap or leave unintended gaps.
4. The admin saves the rules.

Alternative flows:
- Invalid overlapping or gapped ranges are rejected.
- The sample GPA ranges can be changed without code changes.

Postcondition: The system uses the saved configurable rules for future registration checks.

UC-16 - View Registration Records

Primary actor: Admin	**Requirements:** FR-ADM-18

Precondition: The admin is signed in and authorized.

Main flow:
1. The admin views registrations by section or semester.
2. The admin can search for a student's registrations.
3. The system displays the matching registration records.

Alternative flows:
- If no records match, the system shows an empty result.

Postcondition: The admin can review registration data without changing its history.

UC-17 - Validate Registration Eligibility

Primary actor: System (included by Register Course)	**Requirements:** FR-VAL-01 to 09; BR-01 to 06, 10, 11

Precondition: A student has confirmed a course-registration request.

Main flow:
1. The backend checks that the course is offered and the registration period is open.
2. It checks prerequisites, duplicate registration, available section seats, GPA-based credit limit, and lecture-time conflicts.
3. It blocks any request that fails a check and returns a clear reason.

Alternative flows:
- If all checks pass, the system returns an eligible result to Register Course.

Postcondition: Registration eligibility is decided by backend rules before saving.

UC-18 - Check Drop Period

Primary actor: System (included by Drop Course)	**Requirements:** FR-REG-05; BR-07

Precondition: A student has requested to drop a registered course.

Main flow:
1. The system compares the current date with the permitted drop period.
2. It allows the drop only when the period is open.

Alternative flows:
- Outside the period, the system rejects the request and explains why.

Postcondition: The drop decision follows the configured date window.

5. Test Cases Derived from Use Case Flows

These are manual acceptance-level tests. Each expected result is based on the SRS flow or business rule. Sample GPA thresholds are tested exactly as listed in the SRS and should remain configurable.

ID	Scenario and steps	Expected result	SRS trace
TC-01	**Create a valid student account** Submit valid student sign-up information.	A student account is created and confirmation is shown.	FR-AUTH-01
TC-02	**Log in and log out** Log in as Student, then as Admin; sign out each time.	Each user is authenticated and receives permissions for their role; sign-out ends the session.	FR-AUTH-02, FR-AUTH-04
TC-03	**Block student access to protected data** As a Student, request an admin page and another student's data.	Both requests are denied by the server.	FR-AUTH-05, FR-AUTH-06, BR-08, NFR-02
TC-04	**View dashboard and profile** Open the signed-in student's dashboard.	The listed profile, GPA, course, credit-hour, available-hour, and progress values are shown for that student.	FR-PRO-01 to 07
TC-05	**View plan statuses and locked reason** Open the plan and select a course whose prerequisite is not passed.	Course statuses are shown and the locked course displays the unmet prerequisite reason.	FR-PLAN-01 to 04
TC-06	**Search catalog by name and code** Search once by a course name and once by its code.	Matching current-semester courses appear with course details, prerequisites, sections, times, and remaining seats.	FR-CAT-01 to 06
TC-07	**Register an eligible course** Choose an available course and section, review details, and confirm with all rules satisfied.	Backend records one course, confirms it, and updates the course list, credit hours, history, and schedule.	FR-REG-01 to 04, 06, 08; FR-VAL-09
TC-08	**Reject course not offered** Attempt registration for a course not offered in the current semester.	Request is blocked with a clear reason; no registration is saved.	FR-VAL-01, 08
TC-09	**Reject registration outside the period** Attempt registration before the start date or after the end date.	Request is blocked with a clear reason; no registration is saved.	FR-VAL-02, 08; BR-06
TC-10	**Reject unmet prerequisite** Attempt registration without passing a required prerequisite.	Request is blocked and the prerequisite issue is explained.	FR-VAL-03, 08; BR-01
TC-11	**Reject duplicate course registration** Attempt to register in the same course twice in the semester.	Second request is blocked; the original registration remains unchanged.	FR-VAL-04; BR-04
TC-12	**Reject a full section** Attempt registration in a section with no remaining seats.	Request is blocked and no seat count is exceeded.	FR-VAL-05; BR-03, BR-11; NFR-03
TC-13	**Apply GPA credit-hour thresholds** Check sample GPA boundaries: 1.49, 1.50, 2.00, 2.01, 3.00, and 3.01.	Limits are respectively 12, 15, 15, 18, 18, and 21 hours under the initial sample rules.	FR-CRD-01 to 05
TC-14	**Reject credit-hour over-limit** Try adding a course whose hours would exceed the student's configured limit.	Backend rejects it with a clear reason and keeps the existing total unchanged.	FR-VAL-06; FR-CRD-02 to 04; BR-02

ID	Scenario and steps	Expected result	SRS trace
TC-15	**Reject schedule conflict** Choose a section whose lecture time overlaps another registered course.	Registration is blocked and the time conflict is explained.	FR-VAL-07; FR-SCH-04; BR-05
TC-16	**Handle concurrent last-seat requests** Have two registration requests compete for the final section seat.	At most one request succeeds; section capacity is never exceeded.	BR-11; NFR-03
TC-17	**Drop during the permitted period** Drop an active course while the drop period is open.	Registration status and drop date are recorded; hours and schedule update; history is retained.	FR-REG-05 to 07; FR-SCH-03; BR-07
TC-18	**Reject drop outside the permitted period** Attempt to drop a course after the drop period closes.	Request is rejected with a clear reason; the registration remains active.	FR-REG-05; BR-07
TC-19	**View and refresh weekly schedule** View schedule, then register or drop a course.	Schedule lists course, day, start/end time, and location if available, and reflects the change.	FR-SCH-01 to 03
TC-20	**Calculate academic progress** Compare progress for a student with passed and currently registered courses.	Only passed courses count as earned hours; required, remaining, completed, and progress values display correctly.	FR-PRG-01 to 06
TC-21	**View registration history** Open history for a student with current and dropped courses.	Each record shows course, section, semester, registration date, status, and drop date when applicable.	FR-REG-09
TC-22	**Manage course data safely** Admin adds/edits a course, tries a duplicate code and self-prerequisite, then disables a course with history.	Valid changes save; duplicate/self-prerequisite are rejected; historical registrations remain linked.	FR-ADM-01 to 03, 14, 15; BR-12
TC-23	**Manage study plan** Admin changes plan courses, semester distribution, and required total hours.	Saved plan data is reflected in the system.	FR-ADM-04 to 06
TC-24	**Manage section capacity and times** Admin creates/edits a section, changes times/capacity, closes it, and tries lowering capacity below enrollment.	Valid changes save; lowering below current enrollment is rejected; closed section accepts no new students.	FR-ADM-07 to 10, 16
TC-25	**Manage semester and periods** Admin sets current semester, registration dates, and drop dates.	Saved settings control student registration and drop eligibility.	FR-ADM-11 to 13; BR-06, BR-07
TC-26	**Manage GPA rules** Admin edits GPA-hour ranges with valid, overlapping, and gapped values.	Valid rules save and affect checks; overlapping or unintended gaps are rejected.	FR-CRD-05; FR-ADM-17; NFR-06
TC-27	**View registration records** Admin views by section and semester, then searches a student.	Matching registration records are displayed.	FR-ADM-18

6. Notes and open points

The SRS marks GPA limits as sample rules, leaves the secure Admin account provisioning method open, and requires a decision for section-capacity concurrency control. These points need confirmation before implementation-specific tests can be finalized. Frontend checks may help users, but the SRS requires all registration validations to be enforced by the backend.